



Department of Computer Science & Engineering

Programming with Java

Assignment No. 1

Title: Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors.

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Problem Statement

Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors and demonstrate the implementation through suitable examples.

Learning Objectives

- Understand Java environment setup.
- Learn JDK, JRE and JVM.
- Understand variables, methods and constructors.
- Implement these concepts in Java.

Theory

JDK provides development tools, JRE provides the runtime environment, and JVM executes Java bytecode. Variables store data, methods define program behavior, and constructors initialize objects.

Sample Java Program

```
import java.util.*;

class Student

{

    // Instance variables

    String name;

    int age;

    String course;

    // Default Constructor

    Student() {

        name = "Unknown";

        age = 0;

        course = "Not Assigned";

    }

    // Parameterized Constructor (without using 'this')

    Student(String studentName, int studentAge, String studentCourse) {
```

```

    name = studentName;

    age = studentAge;

    course = studentCourse;
}

// Method to display student details
void displayDetails() {

    System.out.println("Student Name : " + name);

    System.out.println("Age      : " + age);

    System.out.println("Course   : " + course);

}

// Method to check eligibility
void checkEligibility() {

    if (age >= 18) {

        System.out.println("Eligible for College Activities");

    } else {

        System.out.println("Not Eligible for College Activities");

    }

}

public static void main()
{

    // Object using Default Constructor

    Student s1 = new Student();

    System.out.println("----- Student 1 -----");

    s1.displayDetails();


    // Object using Parameterized Constructor

```

```
Student s2 = new Student("Ayush Chaubey", 19, "B.Tech CSE");

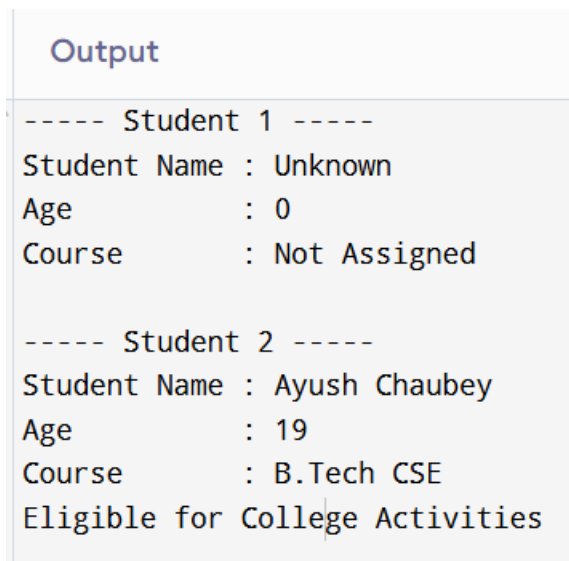
System.out.println("\n----- Student 2 -----");

s2.displayDetails();

s2.checkEligibility();

}
```

Output Screenshot



The screenshot shows the output of a Java program. It displays details for two students. The first student, Student 1, has an unknown name, age 0, and no assigned course. The second student, Student 2, is Ayush Chaubey, 19 years old, studying B.Tech CSE, and is eligible for college activities.

```
Output

----- Student 1 -----
Student Name : Unknown
Age          : 0
Course       : Not Assigned

----- Student 2 -----
Student Name : Ayush Chaubey
Age          : 19
Course       : B.Tech CSE
Eligible for College Activities
```

Conclusion

The assignment demonstrates Java environment setup and the use of JDK, JRE, JVM, variables, methods and constructors.

Questions & Answers

- 1. Explain the main concept used in Java environment setup, variables and methods.

ANS: JDK is used for development, JRE provides the runtime environment, JVM executes bytecode. Variables store data and methods perform tasks.

- 2. What are the advantages of using Java variables and methods?

ANS: They improve code reusability, readability, modularity and maintainability.

- 3. How can this implementation be improved?

ANS: By adding user input, exception handling, comments and advanced OOP concepts.

- 4. What are the real-world applications?

ANS: Enterprise software, Android apps, banking systems, web applications and cloud services.